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NetSages

Explainable Home & Last-Mile Connectivity Diagnosis

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MEASURE
→ **EXPLAIN**
→ **VISUALIZE**

Problem Description

Home internet problems are often described only as “slow Wi-Fi” or “the internet is down.” A speed test may show poor performance, but it rarely explains whether the likely cause is local Wi-Fi, the home router, DNS resolution, the access link, congestion, or a wider destination/path issue.

Key issues addressed

- Users lack a clear, evidence-backed explanation of what a measurement does—and does not—indicate.
- Single snapshots can be misleading; latency, packet loss, throughput and DNS results need context and repeated observations.
- Symptoms from different layers can look similar, making troubleshooting slow and overly dependent on technical expertise.
- Users need practical next checks, not an unsupported claim that the ISP is always responsible.

Goal: convert repeatable connectivity measurements into a transparent diagnosis with evidence, confidence and recommended next steps.

Background

Internet quality is multi-dimensional. Throughput describes successful data transfer rate; latency captures round-trip delay; jitter reflects delay variation; packet loss can disrupt interactive applications; and DNS timing helps distinguish name-resolution delays from other connection problems.

Working model



Scope: a user-space diagnostic prototype for home/last-mile troubleshooting. It cannot directly inspect ISP infrastructure or prove root cause without measurements and network-operator cooperation.

Literature & RFC Review

[1] RFC 6349 — Framework for TCP Throughput Testing

Provides a structured approach to throughput testing and emphasizes baseline RTT, bottleneck bandwidth and test conditions. NetSages uses contextual measurement rather than interpreting a speed result alone.

[2] RFC 9293 — Transmission Control Protocol (TCP)

Defines TCP behavior and gives protocol context for understanding transport performance. TCP throughput can be affected by path conditions and endpoint behavior, so a single low result is not a unique diagnosis.

[3] RFC 7567 — Active Queue Management Recommendations

Explains how queue management and congestion can influence delay and packet loss. Persistent latency increases under load may be a clue, but do not prove a specific router or ISP fault.

[4] RIPE Atlas — Internet measurement platform

Demonstrates the value of repeatable, distributed connectivity measurements and visualizations. NetSages focuses on local user-facing diagnosis and plain-language interpretation.

Research gap: combine multiple measurements with explainable cause hypotheses and user-friendly next actions.

Proposed Solution

NetSages runs a guided diagnostic session, stores time-stamped results, compares them against a local baseline, and generates an evidence-linked explanation. The first version prioritizes transparent rules over opaque prediction: each suggested cause is tied to observed signals and explicit limitations.

1	**1. Capture**	Start a test; record relevant device/network context and consent.
2	**2. Probe**	Measure latency and loss, DNS response time, and throughput where feasible.
3	**3. Normalize**	Repeat tests; summarize median, variability and failures.
4	**4. Diagnose**	Use interpretable rules for Wi-Fi/router, DNS, congestion, access-path or inconclusive cases.
5	**5. Explain**	Show evidence, alternatives, confidence and safe next steps.

Output: diagnosis summary • evidence panel • confidence • next check • exportable report

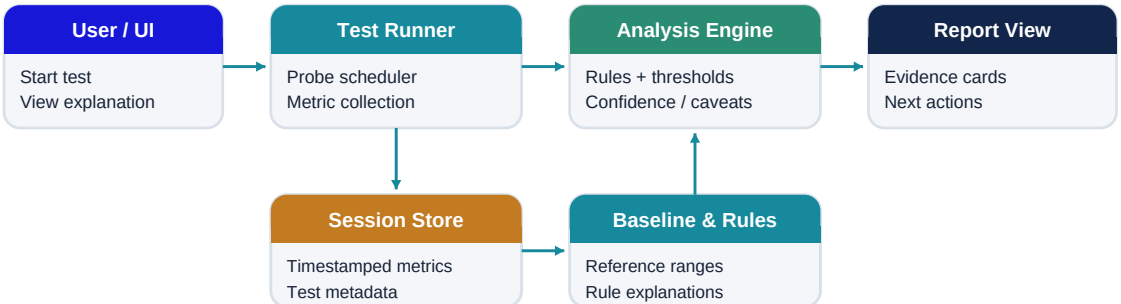
Optimization Proposed

Shift from isolated readings to repeatable, explainable evidence.

Dimension	Basic / before	NetSages / proposed
Measurement	One-off speed test	Repeated latency, loss, DNS and throughput checks
Interpretation	“Slow internet” label	Evidence-linked likely causes with caveats
Reliability	No baseline or context	Baseline comparison, timestamps and variability
User guidance	Generic restart advice	Prioritized checks matched to observed evidence

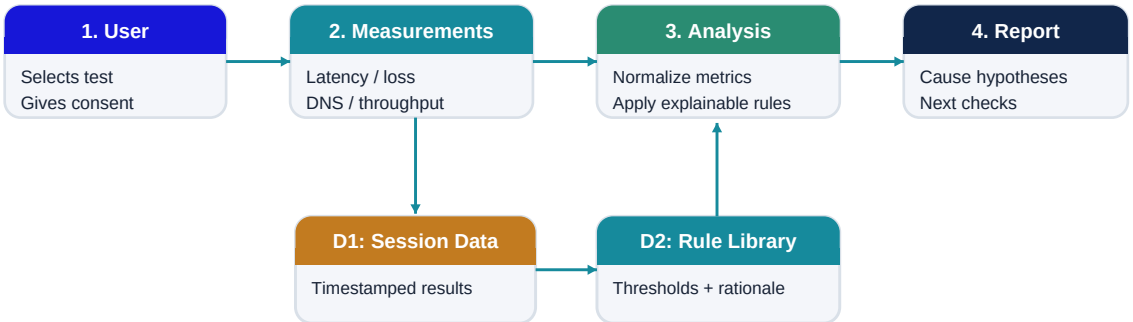
Validate the approach on labeled test scenarios before claiming diagnostic accuracy or performance gains.

Solution Architecture & Design



Logical component view — proposed prototype architecture

Data Flow Diagram



Data handling: collect only what is needed. Avoid passwords, browsing history and payload contents. Treat failed external probes as inconclusive when targets may block them.

Deployment Timeline

Indicative lifecycle for the full project; exact durations depend on scope and testing results.

Phase 1	**Requirements & research**	Define user needs, metrics, scope, constraints and related work.
Phase 2	**Design & planning**	Finalize architecture, data flow, measurement strategy and test cases.
Phase 3	**Core implementation**	Build the measurement runner, session storage and baseline handling.
Phase 4	**Diagnosis & interface**	Implement explainable rules, evidence views and recommended next steps.
Phase 5	**Integration & validation**	Test controlled scenarios; investigate false positives and refine thresholds.
Phase 6	**Deployment & iteration**	Package the demo, document limitations, gather feedback and improve.

Repeat the build–test–review cycle as evidence and project requirements evolve.

Bibliography

[1] B. Constantine, G. Forget, R. Geib, and R. Schrage, “Framework for TCP Throughput Testing,” RFC 6349, Internet Engineering Task Force, Aug. 2011. https://www.rfc-editor.org/rfc/rfc6349

[2] W. Eddy, Ed., “Transmission Control Protocol (TCP),” RFC 9293, Internet Engineering Task Force, Aug. 2022. https://www.rfc-editor.org/rfc/rfc9293

[3] F. Baker and G. Fairhurst, “IETF Recommendations Regarding Active Queue Management,” RFC 7567, Internet Engineering Task Force, July 2015. https://www.rfc-editor.org/rfc/rfc7567

[4] RIPE Network Coordination Centre, “RIPE Atlas: Internet measurements.” https://www.ripe.net/analyse/internet-measurements/ripe-atlas/

These sources inform the networking background. Diagnostic rules and performance claims must be validated experimentally before being presented as results.

Thank You

NetSages — Don’t just measure your internet. Understand it.

MEASURE • EXPLAIN • VISUALIZE